

Module 1

Signal & Systems

Solutions

Question	Final Answer
Q3(a)	Continuous, Deterministic, Periodic
Q3(b)	Discrete, Deterministic, Periodic
Q3(c)	Discrete, Random, Aperiodic
Q3(d)	Continuous, Random, Aperiodic
Q4(a)	Nonlinear, Time-invariant
Q4(b)	Linear, Time-invariant
Q4(c)	Linear, Time-varying
Q4(d)	Linear, Time-invariant
Q5(a)	Causal, Memoryless
Q5(b)	Non-causal, Memoryless
Q5(c)	Causal, With memory
Q5(d)	Causal, With memory
Q6	Energy = 18 , Power = 0
Q7	Energy = 0.5 , Power = 0
Q8	Energy = ∞ , Average Power = 12.5
Q9(a)	Energy signal
Q9(b)	Power signal
Q9(c)	Energy signal (Energy = 1)

Q10

Even component = **0**, Odd component = **$t^3 + t$**